Machine Learning - Graduate Level Assessment 16 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. Which of the following are the spatial clustering algorithms?
   (a) Partitioning based clustering
   (b) K-means clustering
   (c) Grid based clustering
   (d) All of the above

2. Another term for out-of-distribution detection is?
   (a) anomaly detection
   (b) one-class detection
   (c) train-test mismatch robustness
   (d) background detection

3. Which of the following points would Bayesians and frequentists disagree on?
   (a) The use of a non-Gaussian noise model in probabilistic regression.
   (b) The use of probabilistic modelling for regression.
   (c) The use of prior distributions on the parameters in a probabilistic model.
   (d) The use of class priors in Gaussian Discriminant Analysis.

4. Which of the following best describes the joint probability distribution P(X, Y, Z) for the given Bayes net. X <- Y -> Z?
   (a) P(X, Y, Z) = P(Y) * P(X|Y) * P(Z|Y)
   (b) P(X, Y, Z) = P(X) * P(Y|X) * P(Z|Y)
   (c) P(X, Y, Z) = P(Z) * P(X|Z) * P(Y|Z)
   (d) P(X, Y, Z) = P(X) * P(Y) * P(Z)

5. Which of the following best describes what discriminative approaches try to model? (w are the parameters in the model)
   (a) p(y|x, w)
   (b) p(y, x)
   (c) p(w|x, w)
   (d) None of the above

True / False (10 marks)

Answer True or False

6. True or False: Clustering is the best approach for predicting the amount of rainfall based on various cues in meteorological data.

7. True or False: The rank of the matrix A = [[1, 1, 1], [1, 1, 1], [1, 1, 1]] is 3 because all rows are linearly independent.

8. True or False: The Penn Treebank is primarily designed for sentiment analysis rather than language modeling tasks.

9. True or False: The dimensionality of the null space of the matrix A is 3.

10. True or False: In the given Bayesian network, a total of 15 independent parameters are required if no assumptions about independence are made.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to extract maximum and minimum k elements in the given tuple.

12. Write a function to sort a given list of elements in ascending order using heap queue algorithm.

End of Paper